Foundations of Generative AI (ICT4403)

Assignment 2: Autoregressive Text Generation

Student name: Hisham Adil

Registration number: 230953434

System title: Comparative Word-Level Autoregressive Text Generation using Simple RNN, Bidirectional RNN, LSTM and GRU

Dataset: Alice's Adventures in Wonderland by Lewis Carroll

Dataset source: https://www.gutenberg.org/files/11/11-0.txt

GitHub repository: https://github.com/Hisham-byte/RNNTextGeneration

Verified results

Model	Loss	Perplexity	Accuracy	Time (s)
GRU	4.7548	116.1440	18.92%	74.32
SimpleRNN	4.8733	130.7514	18.20%	31.07
LSTM	4.9211	137.1574	17.91%	101.61
Bidirectional RNN	5.0459	155.3812	16.66%	42.69

Conclusion

GRU is the best overall model because it achieved the lowest loss and perplexity and the highest accuracy. Temperature 0.5 was most coherent but repetitive, 1.0 balanced coherence and diversity, and 1.5 increased diversity with reduced grammatical consistency.